

Ayush Agarwal

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Summary!

Multidisciplinary Data Scientist and Software Engineer with a strong foundation in AI/ML, predictive modeling, and full-stack development. Proficient in Python, Java, and React, with proven experience building end-to-end data products—from geospatial data engineering to deploying interactive ROI dashboards. Adept at combining technical depth in algorithms (XGBoost, Deep Learning) with strategic business analysis to deliver scalable, actionable solutions.

Education

Vellore Institute of Technology(Chennai Campus), B.Tech in Computer Science and Engineering with specialization in AI & ML Chennai, India
Sept 2022 – Present

- CGPA: 8.72

Narayana E-Techno School, 12th Mumbai, India
May 2020 – May 2022

- Grade: 88%

RBK School, 10th Mumbai, India
Sept 2022 – Present

- Grade: 94%

Certifications

Artificial Intelligence using TensorFlow, SmartInternz / Google Jun 2024 – Jul 2024

Skills & Interests

Programming Languages: Python, Java, JavaScript, HTML/CSS, PostgreSQL (Complex Joins, Sub-queries)

Frameworks & Libraries: pandas, numpy, scikit-learn, matplotlib, seaborn, YOLOv8, React, Streamlit

Tools & Cloud: AWS, Streamlit Cloud, Tableau, Jupyter, Git/GitHub, VS Code

Core Competencies: ETL Pipelines, Machine Learning (Regression/Classification), A/B Testing, Hypothesis Testing, REST APIs, Agile SDLC

Soft Skills: Problem Solving, Research Documentation, Team Collaboration, Technical Communication

Projects

InsightIQ: Autonomous Text-to-SQL Agent GitHub | Live Demo

- Architected a RAG-based BI agent using Gemini and ChromaDB to translate natural language into **complex SQL queries (JOINS, Sub-queries, Aggregations)**; implemented an agentic self-correction loop for syntax validation.
- Deployed a full-stack dashboard on **Streamlit Cloud** with Role-Based Access Control (RBAC), reducing ad-hoc data retrieval time by 80%.
- Tools Used: Python, Gemini, LangChain, RAG, ChromaDB, PostgreSQL, Streamlit

SCM-Optima: AI Supply Chain Control Tower GitHub | Live Demo

- Developed an analytics platform processing 180,000+ orders; engineered a Random Forest classifier (83% accuracy) to predict shipment delays, validated through comprehensive **Model Evaluation metrics** (Precision, Recall).
- Designed a dynamic inventory optimization engine using Prophet for 30-day demand forecasting and automated Safety Stock calculations.
- Tools Used: Python, Streamlit, Prophet, Scikit-Learn, Plotly, Pandas

VCheck: Uniform Compliance System

- Engineered an AI model using YOLOv8 to detect uniform compliance from images with 98% accuracy; trained on a custom dataset to automate visual inspection workflows.

NYC Rental Arbitrage Engine

GitHub | Live Demo

- Engineered an automated **ETL pipeline** fusing 11,000+ Airbnb listings with Zillow housing data using Geopandas; built an XGBoost price prediction model ($R^2 = 0.51$) interpreted via SHAP analysis.
- Designed a Cash-on-Cash ROI calculator identifying >8% yield opportunities and deployed an interactive geospatial dashboard for real-time investment filtering.
- Tools Used: Python, Geopandas, XGBoost, SHAP, ETL, Streamlit

Research Experience

Predictive Modelling of Physicochemical Properties of Cissus Quadrangularis Compounds *Letters in Applied NanoBioScience*

- Led the end-to-end data pipeline, from scraping chemical property data to training multiple regression models (Random Forest, AdaBoost, Linear) using **Scikit-Learn**.
- Successfully modeled compound behaviors and published results (DOI: 10.33263/LIANBS143.192).

Quantitative Structure-Property Relationship (QSPR) Analysis of Cissus Quadrangularis Compounds Using Topological Indices and Machine Learning Models *Journal of Healthcare Informatics Research (Under Review)*

- Applied Machine Learning algorithms to predict chemical properties using topological descriptors.
- Engineered features based on molecular structure graph theory to improve model interpretability and accuracy.

Modelling of Physicochemical Properties of Papaya Leaf Compounds Using Topological Indices and Random Forest Regression *Journal of Applied Statistics (Submitted)*

- Developed predictive models using topological indices and **Random Forest Regression** to estimate complex compound properties.
- Managed the full lifecycle of the project, including dataset creation, feature selection, and model validation.

Predictive QSAR Modelling of Bioactivity for Phytochemicals from Carica papaya: A Comparative Analysis of Machine Learning Regression Algorithms on a Pilot Dataset *Polycyclic Aromatic Compounds (Submitted)*

- Trained and evaluated a comparative suite of Machine Learning Regression algorithms on a pilot dataset to predict bioactivity.
- Optimized model hyperparameters to minimize error rates in predicting phytochemical behavior.

Proposing the Urban Plant Suitability Index (UPSI): A Multi-Criteria Framework for Selecting Flora in Tropical Urban Ecosystems *Forest Ecology and Management (Submitted)*

- Proposed and designed the **UPSI**, a multi-criteria decision-making framework to assist in urban planning and forestry
- Integrated diverse environmental data points to create a scoring system for selecting flora in tropical ecosystems.

Experience

Web Developer Intern, Afame Technologies — Remote

Apr 2024 – May 2024

- Developed and optimized dynamic, responsive web interfaces using HTML, CSS, and JavaScript, improving load times and user experience for client websites.